HYBRID FOURIER NEURAL OPERATORS FOR TIME-DOMAIN BLACK-HOLE PERTURBATIONS AND QUASINORMAL MODE EXTRACTION

Executive Summary

Jonathan Chung | MPhil Data Intensive Science | University of Cambridge

Motivation & Scientific Justification

A perturbed black hole settles by emitting a ringdown signal composed of damped oscillations, the quasinormal modes (QNMs). For each mode of a Kerr black hole, general relativity fixes the frequency and damping time through the remnant mass and spin [1,2]. Measuring these tones in gravitational-wave data tests whether the remnant is consistent with the Kerr prediction, an observational programme opened by the ringdown analysis of GW150914 [3]. Reliable spectroscopy requires an accurate time-domain waveform and a fit interval in which the QNM model is valid [4].

Physics-informed neural networks (PINNs) approximate differential equations by minimising their governing-equation, initial-data and boundary residuals. Patel et al. [5] solve the Regge–Wheeler and Zerilli equations in the time domain, but report relative L^2 field errors of 23.59% and 28.06%, respectively, and select one fit window manually. We first reproduce and improve this calculation. We then introduce a Schwarzschild hybrid Fourier neural operator (hFNO) that corrects a reduced-resolution numerical prior across mass and initial-data parameters. A single trained hFNO predicts fields for new configurations without per-configuration neural-network retraining.

Methodology

The reproduction uses Patel et al.'s network architecture, loss weights and optimiser schedule. An enhanced PINN then doubles the L-BFGS optimisation budget and adds residual-greedy sampling, placing more training points where the equation residual is largest.

For the Schwarzschild hFNO, Richardson extrapolation of the $k = 2$ and $k = 4$ coarse fields produces a field estimate whose difference from the deployment prior defines the correction target [6]. At inference, the reconstruction is

$$\Phi^{\mathrm{hyb}}(x,t) = \mathcal{U}\Phi^{\mathrm{c}}(x,t) + g(x,t)\,\mathcal{H}_\theta(X)(x,t). \quad (1)$$

Here $\mathcal{U}$ denotes quintic upsampling, Φ^{c} the deployed $k = 4$ coarse field, $\mathcal{H}_\theta$ the four-layer FNO correction operator [7–9], and g the deterministic fixed-scale gradient gate derived from the numerical prior. The fine FD reference field is used only for evaluation and does not enter the training loss. Coarsening both space and time by four reduces the finite-difference point-update count to approximately one-sixteenth of the fine-grid evolution. Each new configuration requires one $k = 4$ evolution, quintic upsampling and one neural-operator evaluation.

An independently trained Kerr hFNO corrects the complex spin--2 Teukolsky field evolved on a horizon-penetrating hyperboloidal domain [10,11]. Its correction target is constructed from independently evolved $N = 201$ and $N = 401$ fourth-order fields; the $N = 801$ fine reference field is used only for evaluation. Unlike the Schwarzschild model, the Kerr correction is ungated because a gradient gate would suppress corrections in the low-amplitude ringdown at future null infinity. QNMs are extracted from the Schwarzschild observer waveform at $x_q = 10M$ and the Kerr waveform at future null infinity. For Schwarzschild, fit-window scans identify stable contiguous blocks of frequency and damping-time estimates. For Kerr, real-component and complex-field estimators generate candidate poles, and the spin-dependent Leaver pole identifies the labelled $(4, 2, 0)$ mode without replacing the fitted values.

Key Findings

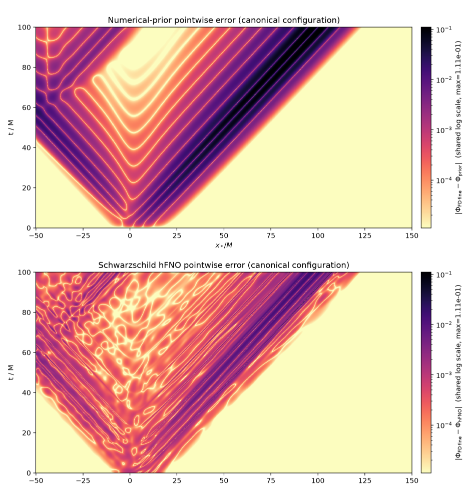

Figure 1: Pointwise Schwarzschild field error on the canonical configuration for the numerical prior (top) and hFNO (bottom), shown on the same logarithmic scale. The hFNO substantially reduces the coherent numerical-prior error bands, while structured wavefront and late-time errors remain.

Our reproduction obtains relative L^2 field errors of 2.61% for Zerilli and 2.67% for Regge–Wheeler, compared with the published 28.06% and 23.59%. The enhanced PINN reaches 0.58% relative L^2 field error. From its observer waveform, the M4 two-mode fit-window plateau estimate gives a 0.029% fundamental-frequency error, while the M2 single-mode nonlinear least-squares fit gives a 1.99% damping-time error relative to Leaver's continued-fraction values [12].

Across 100 independent Schwarzschild configurations, the hFNO reduces the mean relative L^2 error from 13.39% for the numerical prior to 1.85%, and the mean-squared field error by a factor of 46. M4 gives a median fundamental QNM frequency error of 0.152%. Figure 1 shows that the hFNO substantially reduces the coherent wavefront-error bands of the numerical prior, while structured wavefront and late-time errors remain.

Across 128 independent Kerr configurations spanning $a/M \in [0, 0.95]$, the independently trained hFNO reduces the median relative L^2 error from 59.5% for the numerical prior to 0.78%. Under retrospective Leaver-closest reporting, the candidate estimate with the smallest Leaver error is selected separately for frequency and damping time after extraction. The resulting median errors are 0.33% in frequency and 1.52% in damping time, compared with 0.05% and 0.27% for the fine reference field. Complete per-estimator results show substantial method dependence.

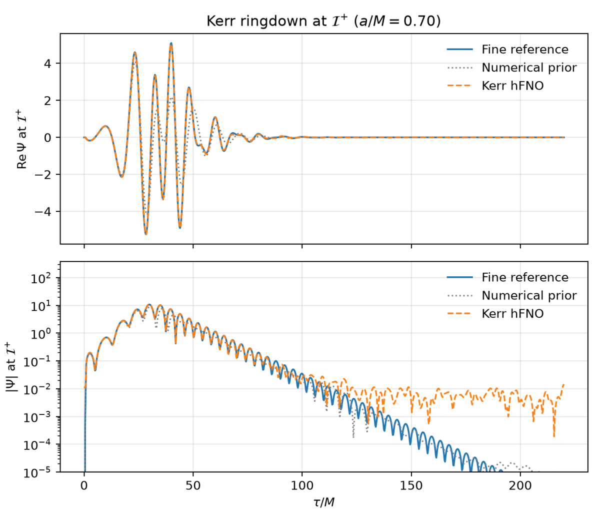

Figure 2: Kerr ringdown at future null infinity for a test configuration nearest $a/M = 0.70$. The Kerr hFNO follows the fine reference through the prompt response and early ringdown, whereas the numerical prior is visibly under-resolved. The logarithmic panel reveals the hFNO's late-time residual, which limits damping-time recovery.

Recommendation & Next Steps

The first priority is a late-time-sensitive training objective that improves the low-amplitude Kerr ringdown. Further tests should cover additional harmonics and initial data, parameters outside the training ranges, and changes of numerical domain and resolution. End-to-end timing should quantify the wall-clock reduction and training break-even point, including numerical evolution, interpolation and neural inference. The resulting waveform families can then be combined with reduced-order likelihood methods to test their practical value for Bayesian parameter estimation [13].

Research Impact & Conclusion

The separately trained Schwarzschild and Kerr hFNOs each combine a reduced-resolution numerical prior with a correction learned across their respective parameter family. The numerical prior supplies characteristic propagation, potential scattering and boundary treatment, while the neural operator predicts a discretisation-error correction. Neither hFNO uses fine-grid reference fields in its training loss, and neither requires per-configuration neural-network retraining. The Schwarzschild M4 analysis and retrospective Kerr analysis both give sub-percent median fundamental-frequency errors while reducing the required numerical evolution. The hFNOs support repeated field generation across the tested families and can be assessed directly in future gravitational-wave likelihood calculations. The coexistence of 0.78% median global Kerr field error and 1.52% damping-time error is consistent with gridwise MSE giving little weight to the low-amplitude late-time ringdown.

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